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SHORT REPORT

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Identification of a dominant *MYH11* causal variant in chronic intestinal pseudo-obstruction: Results of whole-exome sequencing

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Abstract

Chronic Intestinal Pseudo-Obstruction (CIPO) is a rare gastrointestinal disorder, which affects the smooth muscle contractions of the gastrointestinal tract. Dominant mutations in the smooth muscle actin gene, *ACTG2*, accounts for 44%-50% of CIPO patients. Other recessive or X-linked genes, including *MYLK*, *LMOD1*, *RAD21*, *MYH11*, *MYL9*, and *FLNA* were reported in single cases. In this study, we used Whole-Exome Sequencing (WES) to study 23 independent CIPO families including one extended family with 13 affected members. A dominantly inherited rare mutation, c.5819delC (p.Pro1940HisfsTer91), in the smooth muscle myosin gene, *MYH11*, was found in the extended family, shared by 7 affected family members but not by 3 unaffected family members with available DNA, suggesting a high probability of genetic linkage. Gene burden analysis indicates that additional genes, *COL4A1*, *FBLN1* and *HK2*, may be associated with the disease. This study expanded our understanding of CIPO etiology and provided additional genetic evidence to physicians and genetic counselors for CIPO diagnosis.

KEYWORDS
Chronic Intestinal Pseudo-Obstruction, exome-sequencing, genetics, *MYH11*

1 | INTRODUCTION

Chronic Intestinal Pseudo-Obstruction (CIPO) is a progressive disorder with clinical signs varying from mild (constipation) to total disability and death.[1] CIPO can be either myopathic or neuropathic, depending on whether the abnormalities occur in muscles or nerves within the gastrointestinal (GI) tract.[2] The GI symptoms and signs which predominate result from gastroparesis, and include abdominal distension, pain, nausea and vomiting. There is a frequent need for gastrostomy, ileostomy, intestinal resection and/or colectomy. Total parenteral nutrition (TPN) may be needed for decades. In such cases, intestinal failure may eventually lead to total intestinal, liver, pancreas and spleen transplantation. Bladder involvement resulting in megacystis and megaureter may be evident at birth, and even if not, may lead to chronic bladder problems, and/or a life-time of self-catheterization.

The incidence of CIPO in the general population remains unclear due to considerable phenotypic heterogeneity. A Japanese study suggests a prevalence estimation of 1.00 and 0.80 cases per 100 000 males and females for those with severe disease.[3] Autosomal dominant mutations in the smooth muscle actin gene, *ACTG2*, accounts for 44%-50% of CIPO patients.[4] *ACTG2* encodes the smooth muscle γ-2, which is predominantly expressed in enteric tissue. It is one of the contractile apparatus components and pathogenic variants in the protein impair its polymerization and result in the reduced cellular contractility.[5] Additionally, homozygous mutations in *MYLK*,[6] *MYH11*,[7-9]

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LMOD1,[10] *MYL9*,[11] and *RAD21*[12] and X-linked mutation in *FLNA*[13] have been found in a small number of cases. Moreover, compound-heterozygous mutations in *MYH11* have also been reported in two CIPO families.[8,9] The genetic etiology of the remaining CIPO cases, however, remains elusive.

Here, we performed the largest WES study on 23 independent CIPO families with 31 affected patients to identify genetic factors that contribute to CIPO pathogenesis. We report the first description of a dominant pathogenic variant in *MYH11* in a family with CIPO and identify additional genes that may be associated with CIPO.

2 | RESULTS

2.1 | Cohort characteristics

We exome-sequenced 23 independent CIPO families including 20 singletons, 2 duos who are first-degree relatives, and 1 extended family with 7 affected individuals (Supplementary Table 1). They age from birth to 70 years and two patients have died. 12 (38.7%) of them are males and 19 (61.3%) are females. Eleven underwent colectomy, ileostomy, gastrostomy or cholecystectomy. Ten were TPN dependent, 12 had megacystis, 2 had megaduodenum and 5 were subject to total intestinal transplantation. All cases have been prescreened and are negative for known pathogenic variants in *ACTG2*.

2.2 | Damaging heterozygous *MYH11* variant segregated in an extended CIPO family

When screening for LoF or missense variants in known genes causing CIPO, we found that patient 122 057 from family RQ6654 carries a heterozygous frameshift variant (NM_001040113: c.5819delC [p.Pro1940HisfsTer91]) in myosin heavy chain gene *MYH11*. Patient 122 057 has a family history of CIPO and 13 members in the family were diagnosed with CIPO, mostly with megacystis requiring long term self-catheterization (Table 1).

DNA samples were available from 10 family members including 7 affected individuals and 3 unaffected individuals. *MYH11* c.5819delC (p.Pro1940HisfsTer91) was shared by all 7 affected individuals through WES. Further sanger sequencing validation revealed the absence of the variant in 3 unaffected family members, suggesting a high probability of genetic linkage with complete penetrance (LOD = 2.4) (Figure 1, Supplementary Table 2). The variant is rare with MAF = 6.53×10^{-5} in gnomAD WES and absent from gnomAD WGS. It locates in the last exon at the end of the transcript (five amino acids away from the stop codon) and generates a new stop codon 91 amino acids away from the variant (Supplementary Figure 1). It is classified as pathogenic based on ACMG guideline[14] (PVS1, PM2, PP1, and PP2). There are another two rare protein altering variants (MAF $\leq1\times10^{-3}$) in *EARS2* and *TNRC6A* shared by all 7 affected individuals but they are predicted as tolerated by MetaSVM and other 6 commonly used deleteriousness prediction tools (Supplementary Table 3).

2.3 | Other damaging variants in CIPO cases

Although the cohort has been prescreened for pathogenic *ACTG2* mutations, we included two subjects carrying rare missense (MetaSVM "D") *ACTG2* variants (NM_001615: homozygous c.67G>A [p.Ala23Thr] and heterozygous c.337C>G [p.Pro113Ala]) that are benign [BS2, BS4] and of unknown significance (PM2, PP3) respectively based on ACMG guideline.[14]

In order to identify novel causal genes, a one-tailed binomial test was used to prioritize potential causal genes enriched for rare (MAF $\leq1\times10^{-3}$ in gnomAD) and damaging dominant variants (LoF and missense variants with MetaSVM "D") in 23 unrelated CIPO subjects as described before.[15] Given that CIPO has an association with intestinal smooth muscle contraction and most of the identified causal genes are related with contractile apparatus, we prioritized our analysis to highly expressed genes in intestinal smooth muscle using a published RNA-Seq dataset from mouse intestinal mesenchyme at E14.5[16] and defined the highly expressed intestinal smooth muscle cell genes as the top quartile human orthologs (See Supporting Methods). Through the binomial test (Table 2), *COL4A1*, *FBLN1* and *HK2* show significant enrichment (FDR [False discovery rate] < 0.05 using Benjamini-Hochberg method). Four damaging variants were identified in *COL4A1*, while three in *FBLN* and three in *HK2* (Supplementary Table 4). We were able to obtain the DNA from family members of two (out of four) *COL4A1* variant carriers, all three *FBLN1* variant carriers and none of the *HK2* variant carriers. Examining variant loci in family members identified *COL4A1* c.2624C>G (p.Pro875Arg) variant segregated in family 119 533 while the other one *COL4A1* variant and the three *FBLN1* variants did not segregate (Supplementary Figure 2).

TABLE 1 Clinical information for seven affected Individuals from family RQ6654

Sample ID	Gender	Current age	Megacystis	TPN dependence	Major complications	Other disorders
123 342	M	63	Yes	No	No	No
123 344	M	70	Yes	No	No	Cardiovascular disease
122 057	M	66	Yes	No	Bowel obstruction	Multiple sclerosis
121 911	M	32	Yes	Yes	No	No
123 345	F	40	No	No	Rectal prolapse	Thalassemia major
122 363	M	53	Yes	No	No	No
123 323	F	58	No	No	Malrotation	Renal cancer

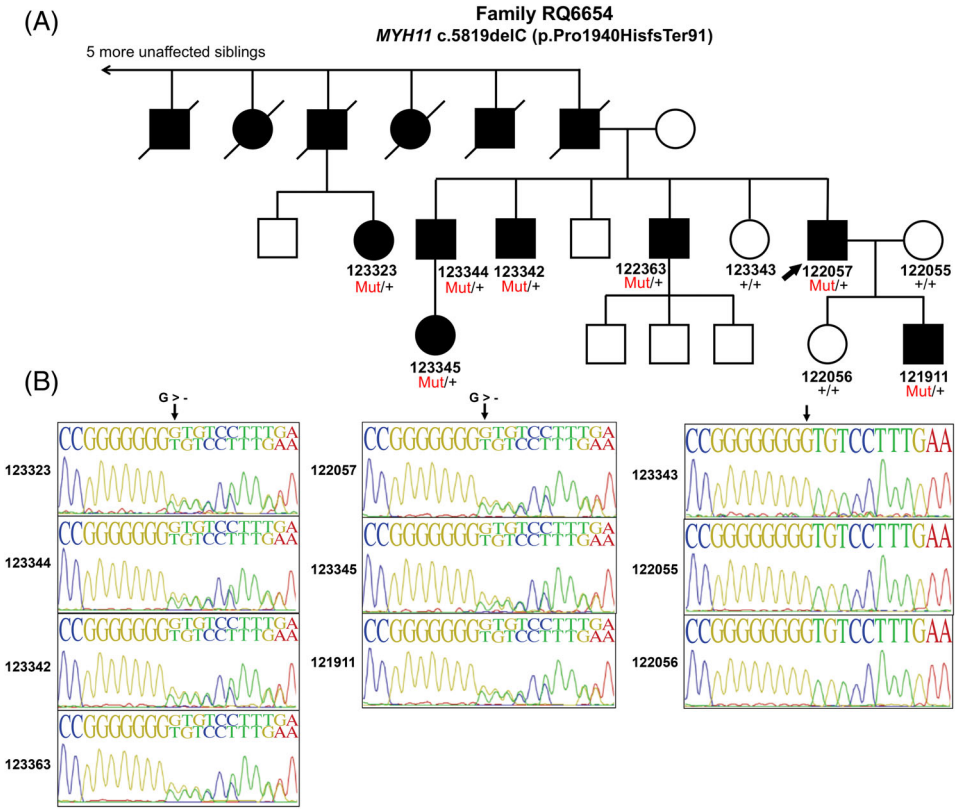

FIGURE 1 *MYH11* variant in the extended CIPO family RQ6654. A, Pedigree of the CIPO family with *MYH11* NM_001040113: c.5819delC (p.Pro1940HisfsTer91) variant. Filled symbols indicate subjects with CIPO; Open symbols represent subjects without CIPO. Arrow indicates the proband. B, Sanger validations in 10 subjects [Colour figure can be viewed at wileyonlinelibrary.com]

TABLE 2 Gene burden analysis for genes that are highly expressed in mouse smooth muscle

Gene	Observed number	Expected number	Fold change	*P*-value	FDR	pLI	Missense Z-score	Smooth muscle expression rank
COL4A1	4	0.10	40	3.89E-06	0.016	1.00	2.46	99.9%
FBLN1	3	0.05	67	1.45E-05	0.030	0.85	1.83	97.8%
HK2	3	0.05	58	2.16E-05	0.030	0.02	2.19	79.1%
SETD1A	3	0.10	29	1.71E-04	0.177	1.00	2.91	94.9%
NDUFS2	2	0.03	72	3.73E-04	0.310	1.00	1.89	93.1%
PYGB	2	0.04	45	9.52E-04	0.660	0.00	0.44	92.8%
TCIRG1	2	0.06	33	1.76E-03	1	0.00	−0.09	83.1%
MYO1C	2	0.08	25	3.11E-03	1	0.00	−0.13	87.0%
LRP5	2	0.09	24	3.40E-03	1	1.00	3.28	94.5%
COL24A1	2	0.10	21	4.24E-03	1	0.00	−2.68	76.2%

Binomial enrichment test was performed for rare damaging (MAF≤1E-03, D-Mis and LoF) variants in top 25% highly expressed smooth muscle genes. Results were ranked by binomial test *P*-value and the top 10 genes were shown. pLI and missense z-score are based on ExAC v0.3.1.

3 | DISCUSSION

We present the largest population-based WES study for CIPO and examined the contribution from potentially damaging exonic variants. The heterozygous frameshift *MYH11* variant c.5819delC (p.Pro1940HisfsTer91) was found with complete segregation in a large family with 7 affected CIPO subjects and 3 unaffected subjects. *MYH11* encodes the smooth muscle myosin heavy chain (SM-MHC) and is highly expressed in the intestinal smooth muscle (Ranked 99.8%). It has four isoforms including SM-A1, SM-A2, SM-B1 and SM-B2. The (+) insert isoforms SM-B1 and SM-B2 are predominantly expressed in small intestine and are responsible for muscle rapid contraction phase.[17] The variant that we found causes stop loss specifically in isoform SM-B2, suggesting a potentially transcript-specific effect. The variant locates at the end of the protein and elongates the protein. Although recessive genotypes in *MYH11* were reported in three separated case (one homozygotes and two compound-heterozygotes) as a cause of megacystis-microcolon-intestinal hypoperistalsis syndrome (Supplementary Figure 1),[7-9] heterozygous *MYH11* pathogenic variants have not been previously reported in CIPO. Heterozygous pathogenic *MYH11* variants

were reported for Thoracic Aortic Aneurysm and/or aortic Dissection (TAAD) and Patent Ductus Arteriosus.[18,19] However, the proband 122 057 and his son have normal aorta through CT scan and MRA, and no family member has had an aneurysm.

MYH11 dimers are an essential component of smooth muscle myosin. One SM-MHC dimer contains two N-terminal globular heads and one C-terminal coiled-coli rod assembly.[20] One study tested the effect of *MYH11* C-terminal variants (NM_002474: c.3743_3814del72 [p.Arg1241_Leu1264del] and c.5361G>A [p.Arg1758Gln]) in TAAD patients, and found the variants reduced the stability of coiled-coil rod and altered its interaction with WT heavy myosin chain.[18] Additionally, both WT and mutant products were expressed, suggesting a dominant negative mechanism.[18] Thus, unlike the homozygous variants found before, the C-terminal frameshift variant in *MYH11* may play a similar dominant negative role that impairs its dimerization and affects muscle contraction.

Additional genes that may play a role in CIPO were identified through gene burden analysis. However, there is limited statistical power to find genes that reach genome-wide significance. Three intestinal smooth muscle highly expressed genes *FBLN1*, *COL4A1* and *HK2* showed the highest significance. Both *FBLN1* and *COL4A1* are related to cellular component organization and extracellular matrix organization, while *HK2* is involved in glucose metabolism. The role of *COL4A1*, *FBLN1* and *HK2* variants in CIPO is unclear. The incomplete segregation observed in some *COL4A1* and *FBLN1* families suggest incomplete penetrance, modifier effect or unrelated to CIPO.

In the majority of subjects with CIPO, however, potentially pathogenic variants were not identified despite the search of the entire exome. This failure may be due to significant genetic heterogeneity of CIPO subjects, restriction to exonic regions, and small cohort size.

This study thus provides insights on future research about CIPO genetic etiology. Firstly, non-coding variants and common variants may also contribute to CIPO pathogenesis and are worth further investigation. Secondly, population-based analysis with large cohort size or family-based analysis may be preferred to increase statistical power. Family studies especially have proved to be an effective way of gene discovery in rare diseases as shown by our identification of a *MYH11* pathogenic variant and previous reports of homozygous variants in consanguineous families.[6,7,10-12] Thus, future studies investigating segregated dominant variants in large CIPO families with multiple affected individuals or homozygotes from consanguineous families may be cost-effective. However, due to the severity and the potential lethality of CIPO, acquiring such families with multiple affected members can also be challenging, as most *ACTG2* pathogenic variants are *de novo* and lead to early death.

In conclusion, this study has expanded the spectrum of genes that are important in the pathogenesis of CIPO. The identification of patients with a mutation in a CIPO gene is important for early clinical treatment and diagnosis. Families with known mutations will have the option of prenatal diagnosis, which has been accomplished,[4] as well as preimplantation genetic diagnosis.

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CONFLICT OF INTEREST

Nothing to declare.

DATA AVAILABILITY STATEMENT

The *MYH11* NM_001040113: c.5819delC (p.Pro1940HisfsTer91) variant and the associated phenotype were submitted to ClinVar database (SUB5617542). Further data of this study are available from the corresponding author upon request.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of this article.

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